# The common logarithm

Lecture and worked examples

# Reference notes

For every $b > 0$, $\log(b)$ is the unique real number $x$ such that $10^x = b$. Hence $\log(10^x) = x$ and, for $b > 0$, $10^{\log(b)} = b$. The function log is defined on $]0, +\infty[$ and it is strictly increasing.

# Reference notes

$\log(a+b)$ does not simplify. Before any logarithmic calculation, check that each argument is strictly positive.

# Reference notes

To solve $a^x = b$, with $a > 0$, $a \neq 1$ and $b > 0$, take the logarithm: $x \log(a) = \log(b)$, so $x = \dfrac{\log(b)}{\log(a)}$.

# Reference notes

Solving $1.08^x = 2$ amounts to writing $x = \dfrac{\log(2)}{\log(1.08)} \approx 9.01$. A value multiplied by 1.08 each year therefore doubles after a little more than nine years.

# Definition of the common logarithm

- Link powers of 10 and logarithms.
- Calculate common values.
- Use the calculator.

# Definition of the common logarithm

Complete the pairs $10^x = b$ and $x = \log(b)$ for $b = 0.001; 0.1; 1; 10; 1{,}000$. Then use the calculator for $b = 7$.

# Definition of the common logarithm

For $b > 0$, $\log(b)$ is the unique solution of $10^x = b$. In particular $\log(1) = 0$, $\log(10) = 1$ and $\log(100) = 2$.

# Definition of the common logarithm

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

# Definition of the common logarithm

**1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

**2 h session:** add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

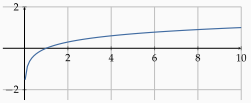
**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

# Variations and graph

- Know the domain.
- Use the fact that the function is increasing.
- Read the curve.

# Variations and graph

Plot $y = 10^x$ and the line $y = x$, then obtain the curve of the logarithm by symmetry. Deduce its domain, its direction of variation and its asymptote.



# Variations and graph

The function log is defined on $]0, +\infty[$, is strictly increasing, and its curve has $x = 0$ as a vertical asymptote.

# Variations and graph

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# Algebraic properties

- Expand or condense logarithms.
- Check the conditions.
- Avoid false properties.

# Algebraic properties

Starting from $10^2 \times 10^3 = 10^5$, conjecture $\log(ab) = \log a + \log b$. Test numerically with $a = 2, b = 5$, then explain why $a, b$ must be positive.

# Algebraic properties

For $a, b > 0$, $\log(ab) = \log a + \log b$ and $\log\left(\frac{a}{b}\right) = \log a - \log b$. Moreover $\log(a^n) = n \log a$.

## Algebraic properties

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# Solving $10^x = b$

- Solve $10^x = b$.
- Bracket a solution.
- Interpret an order of magnitude.

## Solving $10^x = b$

A device measures intensities $b = 0.04; 3.2; 850$. For each value, solve $10^x = b$, give $x$ to the nearest hundredth and check by calculating $10^x$.

## Solving $10^x = b$

The equation $10^x = b$ has a solution only if $b > 0$; this solution is $x = \log(b)$.

## Solving $10^x = b$

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# Solving $a^x = b$

- Solve $a^x = b$.
- Use a quotient of logarithms.
- Round sensibly.

## Solving $a^x = b$

The number of subscribers is multiplied by 1.15 each month. Determine after how many months it will have tripled. Set up the equation, solve it using logarithms and interpret the rounding.

# Solving $a^x = b$

For $a > 0$, $a \neq 1$ and $b > 0$, $a^x = b$ is equivalent to $x = \dfrac{\log(b)}{\log(a)}$.

## Solving $a^x = b$

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# Equations $x^a = b$ and simple inequalities

- Solve equations where the variable is the base.
- Deal with an inequality.
- Take the domain into account.

# Equations $x^a = b$ and simple inequalities

Solve on $[0, +\infty[$: $x^3 = 40$, then $x^4 = 16$. Then solve $1.08^x < 1.5$ and explain the role of the function being increasing.

# Equations $x^a = b$ and simple inequalities

To solve $x^n = b$, take the domain and the parity of $n$ into account. For an exponential inequality, the direction depends on whether the function is increasing or decreasing.

## Equations $x^a = b$ and simple inequalities

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# Problems in context

- Model a situation.
- Choose an equation.
- Conclude in context.

## Problems in context

The sound level is $L = 10 \log\left(\frac{I}{I_0}\right)$. Calculate $L$ when $I = 10^6 I_0$. Then determine the ratio $\frac{I}{I_0}$ for $L = 85$ dB.

## Problems in context

In a problem, define the variable, set up the equation, solve, check the domain and conclude with the unit. A numerical solution without interpretation is incomplete.

## Problems in context

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# Review and assessment

- Summarise the methods.
- Identify the conditions.
- Correct mistakes.

# Review and assessment

Make a map of methods: domain; properties; equations $10^x = b$; equations $a^x = b$; inequalities. Attach a condition and an example to each branch.

# Review and assessment

Before calculating a logarithm: check positivity. Before a comparison: identify the direction of variation. Before a conclusion: check that the result is plausible.

## Review and assessment

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